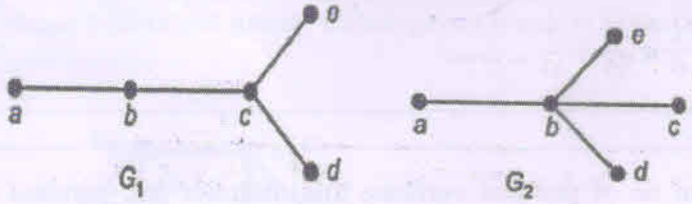
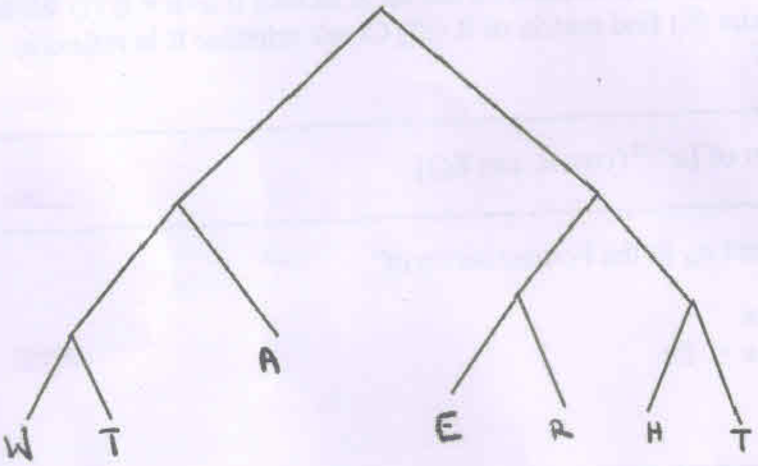
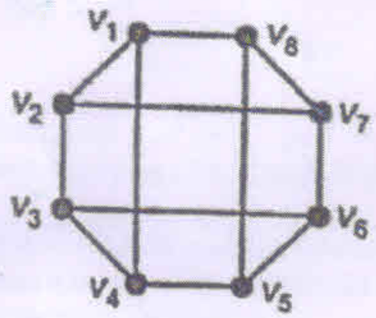
19.12.2022(E)

Maximum Marks: 100	Semester: August 2022 – December 2022 Examination: ESE Examination	Duration:3 Hrs.
Programme code: 04 Programme: B.Tech.	Class: SY	Semester: III (SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: IT
Course Code :116U04C301	Name of the Course: DAM	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	20
i)	Define isomorphic trees Are following trees isomorphic? Check with justification 	5
ii)	Functions $f: R \rightarrow R$, $g: R \rightarrow R$ are defined as $f(x) = 5x + 3$, $g(x) = 1 + 3x$ then find fog, fof, gof & $gogof$	5
iii)	If A={1,2,3,5} and the relation R defined on the set A as aRb if a+b < 8. (i) Write R as a set of ordered pairs (ii) find matrix of R (iii) Check whether R is reflexive, symmetric, transitive ?	5
iv)	Find Laplace Transform of $\{e^{-2t}(\cos 4t\ \cos 8t)\}$	5
v)	Obtain Fourier coefficient a_n in the Fourier series of $f(x) = \begin{cases} x, & 0 < x \leq \pi \\ 2\pi - x, & \pi \leq x \leq 2\pi \end{cases}$ Hence find a_1 & a_2	5
vi)	If 7 points are to be chosen in a regular hexagon of side one unit then show that there are atleast 2 points at a distant less than one unit.	5
Q2 A	Solve the following	10
i)	Find Inverse Laplace Transform of $\left\{\frac{se^{-3s}}{s^2+5s+6}\right\}$	5
ii)	Find Inverse Laplace Transform of $\cot^{-1}(s + 1)$	5
	OR	

Q2 A	Obtain Complex form of Fourier Series for $f(x) = e^{ax}$ *hence* obtain Complex form of Fourier Series for e^{-ax} & $\cosh(ax)$ $in\,(-l, l)$ where a is not an integer.	10
Q 2 B	Solve any **One**	10
i)	Define Heaviside unit step function. Express the following function as Heaviside unit step function and find ~~the~~ Laplace transform $$f(t) = \begin{cases} \cos t, & 0 < t < \pi \\ \cos 2t, & \pi < t < 2\pi \\ \cos 3t, & t > 2\pi \end{cases}$$	10
ii)	Find Fourier series to represent $f(x) = 4 - x^2$ in (0,2) Hence deduce that $\frac{\pi^2}{6} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} \dots\dots\dots$	10
Q3	Solve any **Two**	20
i)	Find total no of edges, total no of pendant vertices, total number non-pendant of vertices, maximum ,minimum height of binary tree with 21 vertices Using the given Huffman tree decode the message (i)00001111100101 (ii)10001101111110 	10
ii)	Using Laplace transform, solve the following differential equation $$\frac{d^2y}{dt^2} + y = t, \quad y(0) = 1, \quad y'(0) = 0$$	10
iii)	If a relation R on the set of integers is defined as a R b if a-b is multiple of 4 Show that the relation is an equivalence relation hence find all equivalence classes	10
Q4.	Solve any **Two**	20
i)	State Convolution theorem and use it to find Inverse Laplace Transform of	10

	$\left\{\frac{s^2}{(s^2+a^2)(s^2+b^2)}\right\}$	
ii)	Prepare table with respect to multiplication in S= Z_7-{$\bar{0}$} Determine whether S is a monoid, a semigroup , a group?	10
iii)	Let A = {a,b,c,d,e}. Let R & S be an equivalence relations on A given as R ={(a,b),(b,a) ,(a,a) ,(b,b) ,(c,c) ,(d,d) ,(e,e) } S= {(a, a) ,(b,b) ,(b,c) ,(c,b) ,(c,c) ,(d,d) ,(e,e)} Write $R^{-1} \cap S^{-1}$, S^{-1} R^{-1} ,SOR find partition of A induced by $R^{-1} \cap S^{-1}$, S^{-1} R^{-1}, SOR	10
Q5	Solve any **four**	**20**
i)	Define a Bipartite graph Check given graph is Bipartite graph or not with justification?  	5
ii)	Show that $f: R-\{1\} \rightarrow R-\{2\}$ such that $f(x)=\frac{2x-3}{x-1}$ is bijective	5
iii)	Let A = {1, 2, 3, 4}, B = {a, b, c, d},C = {x, y, z } and let R = {(1, a), (2, d), (3, a), (3, b), (3, d)} be a relation from A to B and S = {(b, x), (b, z), (c, y), (d, z)} be a relation from B to C. Write SOR Find Domain and Range of SOR	5
iv)	Evaluate $\int_0^{\infty} e^{-3t} t\ \sin t\, dt$	5
v)	Check whether the given function is odd or even & hence decide which fourier coefficients are zero. State the parseval's Identity $$f(x)=\begin{cases}\cos(x), -\pi < x \le 0 \\ cos(\pi - x), 0 \le x \le \pi\end{cases}$$	5
vi)	Define zero divisors and unit elements of Z_n , hence find zero divisors and unit elements of Z_{18}	5

21·12·2022 (E)

SOMAIYA
VIDYAVIHAR UNIVERSITY

Maximum Marks: 100	Semester: August 2022 – December 2022 Examination: ESE Examination	Duration:3 Hrs.
Programme code: 04/02 Programme: B TECH	Class: SY	Semester: III(SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: Information Technology / Computer Engg
Course Code: 116U01C302	Name of the Course: Data Structures	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	20
i)	Define abstract data type. Give example of abstract data type. What is the difference between in built data type and abstract data type?	5
ii)	What are the advantages and disadvantages of circular linked list?	5
iii)	Write applications of priority queue.	5
iv)	Differentiate between B tree and B+ tree.	5
v)	Explain counting sort. Also state the reason why counting sort is called stable sort?	5
vi)	Convert infix expression A*B - C/D + E into prefix.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	What is static memory allocation? Give example of memory allocation. What are the advantages of memory allocation?	5
ii)	What is dynamic memory allocation? Give example of dynamic memory allocation. What are the advantages of dynamic memory allocation?	5
	OR	
Q2 A	What is data structure? How data structure varies from data type? Explain the classification of data structures? What are different types of data structure?	10
Q 2 B	Solve any **One**	10
i)	What is Hashing? Explain the different types of hashing. Which hashing method is better? Justify.	10
ii)	Explain SET as ADT with all operations. Also write application for the same.	10

Que. No.	Question	Max. Marks
Q3	Solve any **Two**	20
i)	What is doubly linked list? Explain working of doubly linked list. Give two real time application of doubly linked list.	10
ii)	Define Stack. Write pseudo code using array or linked list: a. Write a function to check a stack is an underflow b. Write a function to delete an element from stack c. Write a function to display the stack elements	10
iii)	What is Dequeue? Write the pseudo code for insertion at front end and insertion at rear end.	10

Que. No.	Question	Max. Marks
Q4	Solve any **Two**	**20**
i)	Explain the bubble Sort in detail with the help of algorithm. Also sort following numbers 10, 9,11,6,15,2,4,13 using bubble sort by illustrating each step.	10
ii)	Explain the insertion Sort in detail with the pseudo code. Also sort following numbers 23, 34, 2, 43, 15, 25, 65, 51, 44, 8, 21 using insertion sort by illustrating each step.	10
iii)	Explain linear search and binary search with example. Show all the steps.	10

Que. No.	Question	Max. Marks
Q5	Write any **four**	**20**
i)	For web page navigation in forward and backward which data structure will be useful? Justify your answer.	5
ii)	Explain the types of binary tree.	5
iii)	Differentiate between DFS and BFS.	5
iv)	Define MAP ADT and its application.	5
v)	State which data structure is used for BFS traversal and show BFS traversal with an small example	5
vi)	Why linked list is better than array? List different types of linked list?	5

Que. No.	Question	Max. Marks
Q4	Solve any **Two**	**20**
i)	Explain the bubble Sort in detail with the help of algorithm. Also sort following numbers 10, 9,11,6,15,2,4,13 using bubble sort by illustrating each step.	10
ii)	Explain the insertion Sort in detail with the pseudo code. Also sort following numbers 23, 34, 2, 43, 15, 25, 65, 51, 44, 8, 21 using insertion sort by illustrating each step.	10
iii)	Explain linear search and binary search with example. Show all the steps.	10

Que. No.	Question	Max. Marks
Q5	Write any **four**	**20**
i)	For web page navigation in forward and backward which data structure will be useful? Justify your answer.	5
ii)	Explain the types of binary tree.	5
iii)	Differentiate between DFS and BFS.	5
iv)	Define MAP ADT and its application.	5
v)	State which data structure is used for BFS traversal and show BFS traversal with an small example	5
vi)	Why linked list is better than array? List different types of linked list?	5

23.12.2022(E)

SOMAIYA
VIDYAVIHAR UNIVERSITY

Maximum Marks:	**Semester: August 2022 – December 2022** **Examination: ESE Examination**		**Duration:**
Programme code: 03 **Programme: B Tech Information Technology**		**Class: SY**	**Semester: III (SVU 2020)**
Name of the Constituent College: **K. J. Somaiya College of Engineering**		**Name of the department:** IT	
Course Code: 116U04C303	**Name of the Course: Database Management Systems**		
Instructions: 1)Draw neat diagrams 2)Assume suitable data if necessary			

Question No.		Max. Marks
Q1 (a)	Answer any two. i. List significant differences between a file-processing system and a DBMS. ii. Explain the concept of physical data independence, and its importance in database systems iii. What are five main functions of a database administrator?	 5M 5M 5M
Q1 (b)	Suppose you want to build a video site similar to YouTube. Consider each advantages of keeping data in a database management system. Discuss the relevance of each of these points to the storage of actual video data, and to metadata about the video, such as title, the user who uploaded it, tags, and which users viewed it. **OR** Explain the difference between two-tier and three-tier architectures. Which is better suited for Web applications? Why?	10M 10M
Q2 (a)	A university registrar's office maintains data about the following entities: • courses, including number, title, credits, syllabus, and prerequisites; • course offerings, including course number, year, semester, section number, instructor(s), timings, and classroom; • students, including student-id, name, and program; and • Instructors, including identification number, name, department, and title. Further, the enrollment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modeled. i. Construct an E-R diagram that models exams as entities, and uses a ternary relationship, for the above database. ii. Construct appropriate tables for E-R diagram designed with above requirements	10M
Q2 (b)	Consider the following schema: • Suppliers(supplier_number, supplier_name, status, city) • Parts(part_number, part_name, color, weight, city) • Shipments(supplier_number, part_number, quantity) Write SQL query for following requirements. i. Find shipment information (supplier_number, supplier_name, part_number, part_name, quantity) for those having quantity less than	10M

	150. ii. List supplier_number, supplier_name, part_number, part_name for those who made shipment of parts whose quantity is larger than the average quantity. iii. Find aggregate quantity of part_number 'A692' of color 'GREEN' for which shipment made by supplier_number who reside in 'MUMBAI'	
Q3 (a)	Consider following relational database, where the primary keys are underlined. • employee(<u>person_name</u>, street, city) • works(<u>person_name</u>, company_name, salary) • company(<u>company_name</u>, city) • manages(<u>person_name</u>, manager_name) Give an expression in the relational algebra to express each of the following queries: i. Find the names of all employees who live in the same city and on the same street as do their managers. ii. Find the names of all employees in this database who do not work for "First Bank Corporation". iii. Find the names of all employees who earn more than every employee of "Small Bank Corporation".	10M
	OR	
	Solve any two questions. i. Write PL/SQL trigger for following requirement: Event: Deletion of row from stud(roll_no, name, class) table. Action: after deletion of values from stud table, values should be inserted into cancel_admission(roll_no, name) table. (Note: for every row to be deleted, action should be performed.)	5M
	ii. Write a PL/SQL block using user defined exception for following requirements: The Bank_Account table records the current balance for an account, which is updated whenever, any deposit or withdrawals takes place. If the withdrawal attempts is more than the current balance held in the account. The user defined exception is raised, displaying an appropriate message.	5M
	iii. Write a PL/SQL block of code to calculate the factorial value of a number.	5M
Q3 (b)	Normalize the following schema, with given constraints, to 4NF. books(accessionno, isbn, title, author, publisher) users(userid, name, deptid, deptname) accessionno→isbn isbn→title isbn→publisher isbn→→author userid→name userid→deptid deptid→deptname	10M
	OR	
	What is the impact of insert, update and delete anomaly on overall design of database? How normalization is used to remove these anomalies?	10M

<table>
<tr><td>Q4 (a)</td><td>Explain Query processing and Optimization (with appropriate diagram).</td><td>10M</td></tr>
<tr><td>Q4 (b)</td><td>What is hashing? Explain static and dynamic hashing.

OR

When is it preferable to use a dense index rather than a sparse index? Explain your answer. Differentiate clustering index and a secondary index?</td><td>10M

10M</td></tr>
<tr><td>Q5 (a)</td><td>To ensure atomicity despite failure we use recovery methods. Explain in detail Log-Based Recovery method.</td><td>10M</td></tr>
<tr><td>Q5 (b)</td><td>Consider the transaction (T3), Transaction (T4) and Transaction (T6) are any hypothetical transaction working on data item Q. Schedule explaining the execution of T3, T4 and T6 are given below. Decide whether following schedule is conflict serializable or not? Justify your answer.

<table>
<tr><th>T3</th><th>T4</th><th>T6</th></tr>
<tr><td>read (Q)</td><td></td><td></td></tr>
<tr><td></td><td>Write (Q)</td><td></td></tr>
<tr><td>Write (Q)</td><td></td><td></td></tr>
<tr><td></td><td></td><td>Write (Q)</td></tr>
</table>

OR
Transaction during its execution should be in one of the different states at any point of time, explain the different states of transactions during its execution.</td><td>10M

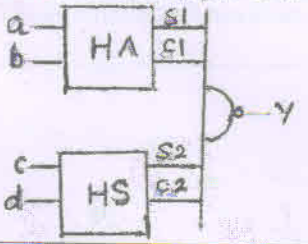
10M</td></tr>
</table>

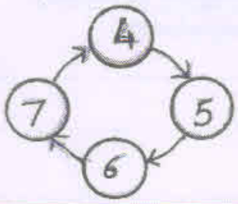
26·12·2022(E)

SOMAIYA
VIDYAVIHAR UNIVERSITY

Semester: August 2022 – December 2022		
Maximum Marks: 100	**Examination: ESE Examination**	**Duration:3 Hrs.**
Programme code: 04 **Programme: BTech-IT**	**Class: SY**	**Semester: III (SVU 2020)**
Name of the Constituent College: **K. J. Somaiya College of Engineering**		**Name of the department:** **Information Technology**
Course Code: 116U04C304	**Name of the Course: DIS (Digital Systems)**	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory **3) Assume suitable data wherever necessary**		

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	**20**
i)	Write the expression Y (a, b, c) = a'. (b' + c') in standard **POS** form	5
ii)	Using 2's complement method calculate $(15)_{10}-(16)_{10}$	5
iii)	Convert $(AB)_{16}$ to binary and then to Gray code and then back to binary.	5
iv)	Convert $(25.25)_{10}$ to binary.	5
v)	500 bytes of data are stored in memory starting at address A000H. What will the address of the last byte?	5
vi)	If Y(a, b, c) = a.b + b.c +a.c, then Y = $\sum$m(??)	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	**10**
i)	Write the Boolean expression for the output of a 2:1 multiplexer in terms of inputs and select lines. Draw the circuit using only NOR gates.	5
ii)	Draw the block diagram of a 4 bit adder/subtractor. Briefly explain how subtraction is done.	5
	OR	
Q2 A	A Half Adder (HA) with inputs a and b, and a Half Subtractor (HS) with inputs c and d are connected to a 4 input NAND gate as shown below. Write the truth table for the output Y in terms of a, b, c and d.  	10
Q 2 B	Solve any **One**	**10**
i)	Rewrite the expression F(a, b, c) = a ⊕ b ⊕ c in:- (a) Standard (Canonical) SOP form (b) Standard (Canonical) POS form	10
ii)	Using K map technique find the reduced expression for:- F(a, b, c, d, e) = πM(0, 2, 8, 9, 10, 16, 18, 24, 25, 26)	10

Que. No.	Question	Max. Marks
Q3	Solve any **Two**	**20**
i)	Convert a T flip flop to a JK flip flop. Show all steps clearly.	10
ii)	Draw the circuit of a 4 bit synchronous counter using D flip flop which can work as a Ring counter or a Johnson counter (using an additional control signal S to make the choice).	10
iii)	Design a synchronous up counter using any type of FF, which counts from 4 to 7 repeatedly as shown below. Your design should avoid any lockout condition. 	10

Que. No.	Question	Max. Marks
Q4	Solve any **Two**	**20**
i)	Draw the block diagram of the 8086 CPU architecture with proper labels for all components.	10
ii)	Draw the circuit diagram of a DRAM cell with proper labels and explain its working.	10
iii)	Explain any five addressing modes in the 8086. Give suitable examples using MOV instruction.	10

Que. No.	Question	Max. Marks
Q5	(Write notes / Short question type) on any **four**	**20**
i)	Write a short note on weighted codes and non-weighted codes.	5
ii)	Write a short note on Master slave flip flop	5
iii)	Write a short note on BCD addition with at least two suitable examples.	5
iv)	Compare combinational circuits vs. sequential circuits.	5
v)	Write the names of any ten registers in the 8086.	5
vi)	Draw the memory hierarchy pyramid. What is the relation between speed, size and cost?	5

28.12.2022(E)

SOMAIYA VIDYAVIHAR UNIVERSITY

Maximum Marks: 100	Semester: August 2022 – December 2022 Examination: ESE Examination		Duration: 3 Hrs
Programme code: 04 Programme: IT (BTech)		Class: SY	Semester: III (SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: IT	
Course Code: 116U04C305	Name of the Course: Data Communication and Networking		
Instructions: 1)Draw neat diagrams 2)Assume suitable data if necessary			

Question No.		Max. Marks
Q1 (a)	Compare various network topologies.	5
Q1 (b)	Explain different guided and unguided transmission media.	5
Q1 (c)	Explain connection oriented and connectionless services.	5
Q1 (d)	Explain in short different framing methods.	5
Q2 (a)	Explain the need for Domain Name System (DNS) and describe the protocol functioning.	10
	OR	
Q2 (a)	Explain sockets and socket programming.	10
Q2 (b)	Explain CSMA protocol. How are collisions handled in CSMA/CD.	10
Q3 (a)	Explain how TCP handles error control and flow control.	10
	OR	
Q3 (a)	Discuss in brief different transport service primitives.	10
Q3 (b)	Explain the function of different internetworking components and state the layer in which they work.	10
Q4 (a)	Describe IPV4 header format with diagram.	10
	OR	
Q4 (a)	Explain Classless Inter Domain Routing (CIDR).	10
Q4 (b)	Explain TCP/IP reference model and compare it with OSI reference model.	10
Q5 (a)	Describe error detection and correction code in brief. Explain CRC with an example.	10
	OR	
Q5 (a)	Explain any four functions of Data link Layer with examples.	10
Q5 (b)	Explain the following in short a) SNMP b) FTP	10